

DMG Mori DMU 50 Operator Manual — DMU-14 Site Copy

5-axis vertical machining center. Controller Fanuc 0i-TF. Installed in Aero-Mill-1 cell, Plant 3 (Wichita).

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1. Machine identification

Field	Value
Site machine ID	DMU-14
Backup machine	DMU-22 (same model, same cell)
Make / model	DMG Mori DMU 50 (5-axis VMC)
Controller	Fanuc 0i-TF — see PDM-MAN-FANUC-0iTF-001
Spindle	HSK-A63, 15 kW, 12,000 rpm max
Travel (X × Y × Z)	500 × 450 × 400 mm
Rotary axes	B-axis (table tilt) ±115°, C-axis (rotation) 360°
Primary use	5-axis finish milling of Part 45621-B (Inconel 718 turbine bracket)

2. Operating envelope and process limits

The following machine-imposed limits apply to all work on DMU-14. Process specifications (see [PLM-PS-INC718-0042](#) for Inconel 718) define the tighter operating ranges for specific materials; the values below are the absolute OEM limits and must never be exceeded.

Parameter	OEM machine limit	Notes
Max spindle speed	12,000 rpm	Reduce for tools > 16 mm diameter; follow tool manufacturer's chart.
Max feed rate (linear)	15,000 mm/min	Rapid traverse is faster; this is the cutting feed limit.
Max table load	200 kg	Including fixture.
Coolant pressure	20 bar through-spindle	Required for deep pockets in Inconel 718 to evacuate chips.
Tool-life warning	per tool — Tool Life Management table	Flank-wear monitoring is manual (visual at tool change). The Inconel 718 process spec sets a flank-wear limit VBmax = 0.20 mm; see PLM-PS-INC718-0042 .

For correct cutting parameters on Part 45621-B OP30 (spindle speed, feed rate, depth of cut) refer to work instruction [PLM-WI-45621B-OP30-RevC](#). That document gives the validated combination of parameters within the OEM envelope above.

3. Common alarms (machine-specific overlay)

This section supplements the controller alarm reference ([PDM-MAN-FANUC-01TF-001](#)) with DMU-14-specific guidance.

Alarm	DMU-14-specific cause / context	First action
300 (APC need ZRN)	Most common trigger on DMU-14 is APC battery depletion after a long shutdown, or encoder cable disturbance during spindle service. See service bulletin PDM-SB-DMU14-2024-03 for the spindle-service-related re-homing procedure.	Perform manual reference return on all 5 axes per §7.2 of the controller manual. If recurrent, replace APC battery (located behind the X-axis cover, see fig. 4-2 of the OEM service manual).
EX1024	Tool magazine pocket position lost (after pneumatic interruption).	Run the tool-magazine re-home macro from the maintenance menu.
EX1108	Coolant tank low.	Top up coolant to fill mark; verify refractometer reading per PLM-PS-INC718-0042 .

4. Daily startup checklist

1. Visual inspection: enclosure, doors, way covers.
2. Power up controller; observe boot self-test.

- 3. Reference all axes (X, Y, Z, B, C).
- 4. Warm up: run program `DMU14_WARMUP_15MIN` . Do **not** begin production cuts until warm-up completes — thermal drift is a documented PFMEA cause of out-of-tolerance features (see `QMS-FMEA-45621B-RevA`).
- 5. Check coolant level and concentration (refractometer).
- 6. Verify active tool table against setup sheet for the part to be run.

Document control / sign-off

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